FEDERAL PUBLIC SERVICE COMMISSION

FPSC-STYLE FULL MOCK TEST

Technical Expert / IT-Computer (BS-17)

Paper type	Objective (MCQ) based, single paper
Total questions	100 MCQs (one mark each)
Part-I: English	20 questions = 20 marks
Part-II: Professional Test	80 questions = 80 marks
Negative marking	0.25 marks deducted for every wrong answer
Passing marks	40 out of 100 (40%)
Suggested practice time	120 minutes (self-set; check your roll number slip for the official time)
Contents	Paper, answer sheet, key, explanations, score guide

Prepared for: Ali Haider Abro

Tip: attempt the paper first without looking at the answer key, then check your score on the last page.

Paper Pattern Analysis

The Commission announced (22 August 2025) that all descriptive tests for BS-16 to BS-21 posts under general recruitment are replaced by objective MCQ-based tests. For BS-16 and BS-17 posts this means one paper of 100 marks, a 40% passing threshold, and negative marking of 0.25 per wrong answer. The syllabus split for your post is English 20 marks and Professional Test 80 marks, which this mock test follows exactly.

Part	Topic	Questions	Marks
I	Vocabulary	8	8
I	Grammar Usage	6	6
I	Sentence Structuring	6	6
II	OSI Model and Network Security	10	10
II	Fundamental Programming	5	5
II	Data Types	4	4
II	Program Logic	4	4
II	Basic Algorithms	5	5
II	Computer Networks and Internet	8	8
II	Designing, Solution and Testing	7	7
II	Programming Languages: C++, Visual Basic and Python	14	14
II	Unix / Red Hat Linux Operating System	9	9
II	Multiple Browsers	3	3
II	Search Engine Optimization (SEO)	5	5
II	Web Development	6	6
	TOTAL	**100**	**100**

Note: FPSC publishes the syllabus topics and marks per part, but not a topic-wise question count for each paper. The distribution above is my estimate: more questions for the larger syllabus areas (networking, programming languages, Linux) and fewer for smaller ones (browsers, data types). Question styles follow what FPSC MCQ papers usually ask: direct definitions, "which of the following" choices, short code output questions, synonyms/antonyms, and sentence correction.

Instructions

1. Each question has four options; choose the single best answer.
2. Every correct answer earns 1 mark. Every wrong answer loses 0.25 marks. Unattempted questions score zero.
3. Guess only when you can eliminate at least two options; a blind guess loses marks on average under 0.25 negative marking.
4. Mark your answers on the answer sheet on the next pages, then compare with the answer key.
5. Do not look at the explanations section until you finish the whole test.

PART-I: ENGLISH (20 Marks)

Vocabulary (Q1 - Q8)

1. Choose the word closest in meaning to METICULOUS.
(A) Careless and hasty
(B) Lazy and slow
(C) Angry and impatient
(D) Extremely careful and precise

2. Choose the word OPPOSITE in meaning to SCARCE.
(A) Rare
(B) Limited
(C) Insufficient
(D) Plentiful

3. Choose the word closest in meaning to CANDID.
(A) Frank and honest
(B) Secretive
(C) Hesitant
(D) Impolite

4. What is the meaning of EPHEMERAL?
(A) Lasting forever
(B) Extremely heavy
(C) Lasting for a very short time
(D) Very ancient

5. Choose the word OPPOSITE in meaning to BENEVOLENT.
(A) Kind
(B) Charitable
(C) Generous
(D) Malevolent

6. Choose the correct meaning of the idiom 'to beat around the bush'.
(A) To search carefully for something lost
(B) To attack someone aggressively
(C) To work hard in a garden
(D) To avoid coming to the main point directly

7. Which one word means 'a lover of books'?
(A) Bibliophile
(B) Philatelist
(C) Misanthrope
(D) Numismatist

8. Choose the correctly spelled word.
(A) Acommodation
(B) Accomodation
(C) Accommodation
(D) Acomodation

Grammar Usage (Q9 - Q14)

9. Neither of the students ____ present yesterday.
(A) were
(B) are
(C) was
(D) have been

10. She has been living in Karachi ____ 2015.
(A) for
(B) from
(C) during
(D) since

11. If I ____ you, I would apologize to him.
(A) were
(B) was
(C) am
(D) be

12. He is very good ____ mathematics.
(A) in
(B) on
(C) with
(D) at

13. He is ____ honest man.
(A) a
(B) the
(C) an
(D) no article is needed

14. By the time we reached the station, the train ____.
(A) has left
(B) leaves
(C) had left
(D) was leaving

Sentence Structuring (Q15 - Q20)

15. Arrange the parts to form a meaningful sentence:
P: to school Q: walks R: every day S: Ali
(A) SPQR
(B) QSPR
(C) RSQP
(D) SQPR

16. Find the part of the sentence that has an error:
'Each of the boys have completed their homework.'
(A) Each of the boys
(B) have completed
(C) their homework
(D) No error

17. Choose the grammatically correct sentence.
(A) He does not know the answer.
(B) He do not know the answer.
(C) He does not knows the answer.
(D) He not know the answer.

18. Change into the passive voice:
'The committee approved the proposal.'
(A) The proposal is approved by the committee.
(B) The proposal was approved by the committee.
(C) The proposal has been approving by the committee.
(D) The committee was approved by the proposal.

19. Change into indirect speech:

He said, "I am busy."

(A) He said that I am busy.
(B) He said that he is being busy.
(C) He said that he was busy.
(D) He said that he will be busy.

20. Choose the best improvement for the underlined idea:

'The news are good.'

(A) The news were good.
(B) The news have been good.
(C) The news is good.
(D) The newses are good.

PART-II: PROFESSIONAL TEST (80 Marks)

OSI Model and Network Security (Q21 - Q30)

21. Which layer of the OSI model is responsible for logical addressing and routing?

(A) Network layer
(B) Data Link layer
(C) Transport layer
(D) Session layer

22. Data encryption, compression and translation between data formats are functions of the ____ layer.

(A) Application
(B) Session
(C) Transport
(D) Presentation

23. The Protocol Data Unit (PDU) at the Transport layer for TCP is called a:

(A) Segment
(B) Packet
(C) Frame
(D) Bit

24. Which of the following is a connectionless transport protocol?

(A) TCP
(B) UDP
(C) HTTP
(D) FTP

25. Which device works at the Data Link layer and forwards frames using MAC addresses?

(A) Hub
(B) Repeater
(C) Router
(D) Switch

26. A firewall is primarily used to:

(A) Increase the speed of the internet connection
(B) Filter incoming and outgoing network traffic according to security rules
(C) Take backup of important data
(D) Convert domain names into IP addresses

27. Asymmetric-key cryptography (for example RSA) uses:

(A) A single shared secret key
(B) No key at all
(C) Two identical secret keys
(D) A pair of mathematically related public and private keys

28. An attack in which many compromised computers flood a target with traffic to make it unavailable is called:
(A) Phishing
(B) SQL injection
(C) DDoS attack
(D) Man-in-the-middle attack

29. In the CIA triad of information security, protecting data from unauthorized modification is called:
(A) Confidentiality
(B) Availability
(C) Integrity
(D) Authentication

30. Which of the following is a hashing algorithm?
(A) RSA
(B) AES
(C) SHA-256
(D) DES

Fundamental Programming (Q31 - Q35)

31. A translator that converts the entire source program into machine code before execution is called a:
(A) Compiler
(B) Interpreter
(C) Assembler
(D) Loader

32. An error caused by breaking the grammar rules of a programming language is called a:
(A) Syntax error
(B) Logical error
(C) Runtime error
(D) Design error

33. A variable is best described as:
(A) A fixed value that never changes
(B) A named memory location whose value can change
(C) A keyword reserved by the language
(D) A type of loop

34. Which of the following is NOT a looping construct?
(A) for
(B) while
(C) if-else
(D) do-while

35. Pseudocode is:
(A) The machine code of a program
(B) A graphical diagram of program flow
(C) An informal, language-independent description of an algorithm
(D) The compiled output file

Data Types (Q36 - Q39)

36. On most modern 32-bit and 64-bit systems, the size of an int in C++ is typically:
(A) 1 byte
(B) 4 bytes
(C) 2 bytes
(D) 8 bytes

37. Which data type is used to store only true or false values in C++?

(A) bool
(B) char
(C) float
(D) string

38. In C++, what is the value of 7 / 2 when both operands are integers?

(A) 3.5
(B) 3
(C) 4
(D) 2

39. In C++, what is the value of (int)9.99?

(A) 10
(B) 9.99
(C) 9.9
(D) 9

Program Logic (Q40 - Q43)

40. Which flowchart symbol is used to represent a decision?

(A) Diamond
(B) Rectangle
(C) Oval
(D) Parallelogram

41. What is the result of the logical expression (5 > 3) AND (2 > 4)?

(A) False
(B) True
(C) Error
(D) Cannot be determined

42. The output of an XOR (exclusive OR) gate is true when:

(A) The two inputs are the same
(B) The two inputs are different
(C) Both inputs are 1 only
(D) Both inputs are 0 only

43. What is printed by the following pseudo-code?

```
sum = 0
i = 1
while i <= 4:
    sum = sum + i
    i = i + 1
print(sum)
```

(A) 10
(B) 6
(C) 15
(D) 4

Basic Algorithms (Q44 - Q48)

44. The time complexity of binary search in the worst case is:

(A) O(log n)
(B) O(n)
(C) O(n log n)
(D) O(1)

45. Which sorting algorithm repeatedly compares adjacent elements and swaps them if they are in the wrong order?
(A) Selection sort
(B) Insertion sort
(C) Quick sort
(D) Bubble sort

46. Binary search can be applied only on:
(A) An unsorted list
(B) A list containing only positive numbers
(C) A sorted list
(D) A list with no duplicate values

47. Which data structure follows the LIFO (Last In, First Out) principle?
(A) Queue
(B) Array
(C) Graph
(D) Stack

48. What is the worst-case time complexity of linear search?
(A) O(1)
(B) O(n)
(C) O(log n)
(D) O(n squared)

Computer Networks and Internet (Q49 - Q56)

49. In classful IPv4 addressing, the first-octet range of Class A addresses is:
(A) 128 to 191
(B) 1 to 126
(C) 192 to 223
(D) 224 to 239

50. The Domain Name System (DNS) is used to:
(A) Assign IP addresses to hosts automatically
(B) Encrypt web traffic
(C) Translate domain names into IP addresses
(D) Route packets between networks

51. In a star topology, all devices are connected to:
(A) A single backbone cable
(B) A central hub or switch
(C) Each other in a closed ring
(D) Every other device directly

52. Which protocol automatically assigns IP addresses to hosts on a network?
(A) DNS
(B) DHCP
(C) ARP
(D) ICMP

53. An IPv6 address is how many bits long?
(A) 32 bits
(B) 64 bits
(C) 256 bits
(D) 128 bits

54. Which protocol is used for SENDING e-mail from a client to a mail server?
(A) SMTP
(B) POP3
(C) IMAP
(D) FTP

55. A network that covers a city or a large campus is generally called a:
(A) MAN
(B) LAN
(C) WAN
(D) PAN

56. The subnet mask 255.255.255.0 is written in CIDR notation as:
(A) /8
(B) /24
(C) /16
(D) /32

Designing, Solution and Testing (Q57 - Q63)

57. Which phase comes first in the Software Development Life Cycle (SDLC)?
(A) Coding
(B) Requirement analysis
(C) Testing
(D) Maintenance

58. Which statement correctly describes the Waterfall model?
(A) Each phase must be completed before the next phase begins
(B) Phases overlap and repeat in short cycles
(C) Customer feedback is taken after every sprint
(D) It is best for projects whose requirements keep changing

59. Testing in which the tester has NO knowledge of the internal code structure is called:
(A) White-box testing
(B) Structural testing
(C) Black-box testing
(D) Code coverage testing

60. Testing of individual modules or functions, usually done by developers, is called:
(A) Integration testing
(B) Unit testing
(C) System testing
(D) Acceptance testing

61. Testing performed by end users or clients to decide whether to accept the software is:
(A) Unit testing
(B) Integration testing
(C) User acceptance testing (UAT)
(D) Stress testing

62. Which UML diagram shows the interaction between actors and the functions of a system?
(A) Class diagram
(B) Sequence diagram
(C) Component diagram
(D) Use case diagram

63. The main purpose of regression testing is to:
(A) Confirm that new changes have not broken existing functionality
(B) Test the system under extremely heavy load
(C) Check only the newly added features
(D) Find defects in the requirements document

Programming Languages: C++, Visual Basic and Python (Q64 - Q77)

64. In C++, which operator is used to access a member through a pointer to an object?

(A) . (dot operator)
(B) :: (scope resolution)
(C) -> (arrow operator)
(D) & (address-of)

65. Which of the following is a valid single-line comment in C++?

(A) # This is a comment
(B) // This is a comment
(C) ' This is a comment
(D) <!-- This is a comment -->

66. What is the output of the following C++ code?

```
int x = 5;
cout << x++;
cout << x;
```

(A) 56
(B) 55
(C) 66
(D) 65

67. Wrapping data and the functions that operate on it into a single unit (a class) is called:

(A) Inheritance
(B) Polymorphism
(C) Recursion
(D) Encapsulation

68. Which statement about a C++ constructor is correct?

(A) It has the same name as the class and has no return type
(B) It must be called manually after creating an object
(C) It always returns an int value
(D) It is used to destroy an object

69. Which C++ operator is used to allocate memory dynamically?

(A) delete
(B) sizeof
(C) new
(D) this

70. In Visual Basic, which operator is mainly used to join (concatenate) two strings?

(A) #
(B) %
(C) @
(D) &

71. Which keyword is used to declare a variable in Visual Basic?

(A) Var
(B) Dim
(C) Let
(D) Int

72. Which function displays a dialog box with a message to the user in Visual Basic?

(A) Echo
(B) Cout
(C) Alert
(D) MsgBox

73. What is the output of the following Python code?

```
print(type([]))
```

(A) <class 'list'>
(B) <class 'tuple'>
(C) <class 'dict'>
(D) <class 'array'>

74. Which of the following Python data types is immutable?

(A) list
(B) dictionary
(C) set
(D) tuple

75. Which keyword is used to define a function in Python?

(A) function
(B) fun
(C) def
(D) define

76. What is the output of the following Python code?

```
print(2 ** 3)
```

(A) 6
(B) 9
(C) 5
(D) 8

77. What is the output of the following Python code?

```
print(10 // 3, 10 % 3)
```

(A) 3.33 1
(B) 3 1
(C) 4 1
(D) 3 0

Unix / Red Hat Linux Operating System (Q78 - Q86)

78. Which Linux command lists the files and directories in the current directory?

(A) cd
(B) rm
(C) ls
(D) cat

79. Which Linux command is used to change the permissions of a file?

(A) chown
(B) chgrp
(C) passwd
(D) chmod

80. After running chmod 755 script.sh, the permissions of the file become:

(A) rwxrwxrwx
(B) rwxr-xr-x
(C) rw-r--r--
(D) rwx------

81. Who created the Linux kernel in 1991?

(A) Dennis Ritchie
(B) Linus Torvalds
(C) Richard Stallman
(D) Ken Thompson

82. Which command displays the full path of the current working directory?
(A) pwd
(B) cd
(C) whoami
(D) hostname

83. Red Hat Enterprise Linux and its derivatives mainly use which package format?
(A) .rpm
(B) .deb
(C) .apk
(D) .exe

84. What is the home directory of the root (superuser) account in Linux?
(A) /home
(B) /usr
(C) /root
(D) /etc

85. Which command displays the contents of a file one screen at a time?
(A) cat
(B) more
(C) touch
(D) mkdir

86. The grep command is used to:
(A) Compress files
(B) Change file ownership
(C) Search for text patterns inside files
(D) Show running processes

Multiple Browsers (Q87 - Q89)

87. Which rendering engine is used by Mozilla Firefox?
(A) Blink
(B) Gecko
(C) WebKit
(D) Trident

88. What does the private (incognito) browsing mode mainly do?
(A) It hides your IP address from websites
(B) It makes you anonymous to your internet service provider
(C) It blocks all advertisements
(D) It does not save history, cookies and form data on the device after the session ends

89. Which of the following is NOT a web browser?
(A) Apache
(B) Opera
(C) Safari
(D) Microsoft Edge

Search Engine Optimization (SEO) (Q90 - Q94)

90. The main goal of Search Engine Optimization (SEO) is to:
(A) Increase only the loading speed of a website
(B) Improve a website's visibility in organic (unpaid) search results
(C) Pay search engines for ad placement
(D) Design the graphics of a website

91. Which of the following is an ON-PAGE SEO factor?
(A) Acquiring backlinks from other websites
(B) Optimizing title tags and meta descriptions
(C) Sharing the site on social media
(D) Guest posting on other blogs

92. Backlinks in SEO are:
(A) Links from other websites that point to your website
(B) Links between pages of the same website
(C) Broken links on a web page
(D) Links that open in a new browser tab

93. The robots.txt file is used to:
(A) Store user passwords securely
(B) Tell search engine crawlers which parts of a site they should not crawl
(C) Speed up the website
(D) Create automatic backups

94. Which of the following is an example of black-hat SEO?
(A) Writing original content
(B) Adding alt text to images
(C) Improving page speed
(D) Keyword stuffing

Web Development (Q95 - Q100)

95. In HTML, which tag defines the largest (top-level) heading?
(A) <h6>
(B) <head>
(C) <title>
(D) <h1>

96. Which CSS property is used to change the colour of text?
(A) font-color
(B) text-color
(C) foreground
(D) color

97. Which HTML tag is used to create a hyperlink?
(A) <link>
(B) <href>
(C) <a>
(D) <url>

98. Which of the following is primarily a client-side scripting language?
(A) JavaScript
(B) PHP
(C) ASP
(D) JSP

99. What does the HTTP status code 404 mean?
(A) Forbidden
(B) Internal Server Error
(C) Not Found
(D) OK

100. Which HTTP method is generally used to send form data inside the body of the request?
(A) GET
(B) HEAD
(C) POST
(D) TRACE

Answer Sheet (write your answers here)

Circle one letter per question. Use this sheet to take the test, then check the answer key.

Q	A	B	C	D	Q	A	B	C	D	Q	A	B	C	D	Q	A	B	C	D
1	A	B	C	D	26	A	B	C	D	51	A	B	C	D	76	A	B	C	D
2	A	B	C	D	27	A	B	C	D	52	A	B	C	D	77	A	B	C	D
3	A	B	C	D	28	A	B	C	D	53	A	B	C	D	78	A	B	C	D
4	A	B	C	D	29	A	B	C	D	54	A	B	C	D	79	A	B	C	D
5	A	B	C	D	30	A	B	C	D	55	A	B	C	D	80	A	B	C	D
6	A	B	C	D	31	A	B	C	D	56	A	B	C	D	81	A	B	C	D
7	A	B	C	D	32	A	B	C	D	57	A	B	C	D	82	A	B	C	D
8	A	B	C	D	33	A	B	C	D	58	A	B	C	D	83	A	B	C	D
9	A	B	C	D	34	A	B	C	D	59	A	B	C	D	84	A	B	C	D
10	A	B	C	D	35	A	B	C	D	60	A	B	C	D	85	A	B	C	D
11	A	B	C	D	36	A	B	C	D	61	A	B	C	D	86	A	B	C	D
12	A	B	C	D	37	A	B	C	D	62	A	B	C	D	87	A	B	C	D
13	A	B	C	D	38	A	B	C	D	63	A	B	C	D	88	A	B	C	D
14	A	B	C	D	39	A	B	C	D	64	A	B	C	D	89	A	B	C	D
15	A	B	C	D	40	A	B	C	D	65	A	B	C	D	90	A	B	C	D
16	A	B	C	D	41	A	B	C	D	66	A	B	C	D	91	A	B	C	D
17	A	B	C	D	42	A	B	C	D	67	A	B	C	D	92	A	B	C	D
18	A	B	C	D	43	A	B	C	D	68	A	B	C	D	93	A	B	C	D
19	A	B	C	D	44	A	B	C	D	69	A	B	C	D	94	A	B	C	D
20	A	B	C	D	45	A	B	C	D	70	A	B	C	D	95	A	B	C	D
21	A	B	C	D	46	A	B	C	D	71	A	B	C	D	96	A	B	C	D
22	A	B	C	D	47	A	B	C	D	72	A	B	C	D	97	A	B	C	D
23	A	B	C	D	48	A	B	C	D	73	A	B	C	D	98	A	B	C	D
24	A	B	C	D	49	A	B	C	D	74	A	B	C	D	99	A	B	C	D
25	A	B	C	D	50	A	B	C	D	75	A	B	C	D	100	A	B	C	D

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	21	A	41	A	61	C	81	B
2	D	22	D	42	B	62	D	82	A
3	A	23	A	43	A	63	A	83	A
4	C	24	B	44	A	64	C	84	C
5	D	25	D	45	D	65	B	85	B
6	D	26	B	46	C	66	A	86	C
7	A	27	D	47	D	67	D	87	B
8	C	28	C	48	B	68	A	88	D
9	C	29	C	49	B	69	C	89	A
10	D	30	C	50	C	70	D	90	B
11	A	31	A	51	B	71	B	91	B
12	D	32	A	52	B	72	D	92	A
13	C	33	B	53	D	73	A	93	B
14	C	34	C	54	A	74	D	94	D
15	D	35	C	55	A	75	C	95	D
16	B	36	B	56	B	76	D	96	D
17	A	37	A	57	B	77	B	97	C
18	B	38	B	58	A	78	C	98	A
19	C	39	D	59	C	79	D	99	C
20	C	40	A	60	B	80	B	100	C

Answer distribution: A = 25, B = 25, C = 25, D = 25.

Detailed Answers and Explanations

For each question: the correct answer, why it is correct, why the other options are wrong, and the trap or memory aid to remember.

PART-I: ENGLISH (20 Marks)

Vocabulary

Q1. Correct answer: (D) Extremely careful and precise

Meticulous means paying great attention to detail. Close synonyms are scrupulous, painstaking and thorough. The other options describe the opposite (careless/hasty) or have nothing to do with detail. Trap: FPSC often places the antonym among the options, so confirm whether the question asks for a synonym or an antonym.

Q2. Correct answer: (D) Plentiful

Scarce means in short supply. Rare, limited and insufficient are all synonyms of scarce, so they cannot be antonyms. Plentiful (abundant) is the true opposite. Tip: when three options mean the same thing, the odd one out is usually the answer for an antonym question.

Q3. Correct answer: (A) Frank and honest

Candid means truthful and straightforward in speech. It is not the same as rude, so 'impolite' is a trap. Secretive is nearly its opposite. Remember: candid = frank = open = sincere.

Q4. Correct answer: (C) Lasting for a very short time

Ephemeral comes from a Greek word meaning 'lasting only a day'. Things like morning dew or a mayfly's life are ephemeral. Its opposite is permanent or eternal. Trap: do not confuse it with 'ethereal' (light, delicate, heavenly).

Q5. Correct answer: (D) Malevolent

Benevolent (bene = good) means kind and well-meaning. Malevolent (mal = bad) means wishing harm to others. Kind, charitable and generous are synonyms of benevolent. Memory aid: the Latin roots bene = good, mal = bad.

Q6. Correct answer: (D) To avoid coming to the main point directly

The idiom means talking about unimportant things instead of getting to the point. It has no literal link with plants or fighting. A typical use: 'Stop beating around the bush and tell me what happened.'

Q7. Correct answer: (A) Bibliophile

A bibliophile loves books (biblio = book, phile = lover). A philatelist collects stamps, a numismatist collects coins, and a misanthrope dislikes people. One-word-substitution questions are common in FPSC vocabulary, so learn the '-phile', '-phobia' and '-ologist' families.

Q8. Correct answer: (C) Accommodation

Accommodation has a double c and a double m. Memory aid: the hotel is big enough to hold two Cs and two Ms. Other commonly misspelled words: necessary, occurrence, separate, definitely.

Grammar Usage

Q9. Correct answer: (C) was

'Neither' is singular in standard grammar even when followed by 'of the students', so it takes a singular verb. 'Yesterday' shows past tense, so 'was' is correct. 'Are' and 'have been' do not match the past time. 'Were' is common in speech but is treated as incorrect in formal exams.

Q10. Correct answer: (D) since

'Since' is used with a fixed starting point in time (2015, Monday, last year). 'For' is used with a length of time (for ten years). 'From' and 'during' do not fit the present perfect continuous tense here.

Q11. Correct answer: (A) were

This is the second conditional, which describes an imaginary situation. In formal grammar 'were' is used for all persons after 'if' (If I were you, If he were here). Pattern: If + past simple (were), would + base verb.

Q12. Correct answer: (D) at

The fixed preposition after 'good' (skill) is 'at': good at, bad at, clever at. Preposition-collocation questions must be memorized as pairs: interested in, afraid of, fond of, married to.

Q13. Correct answer: (C) an

The article depends on the SOUND, not the spelling. 'Honest' begins with a silent h, so it sounds like 'onest' and takes 'an'. Compare: an hour, an heir, but a university (sounds like 'yoo'), a European.

Q14. Correct answer: (C) had left

Two past events are described, and the train's departure happened first. The earlier past action uses the past perfect (had + past participle). The signal phrase 'By the time + past simple' almost always needs past perfect in the main clause.

Sentence Structuring

Q15. Correct answer: (D) SQPR

Start with the subject (S: Ali), then the verb (Q: walks), then the place (P: to school), then the time (R: every day), giving 'Ali walks to school every day'. English follows the order Subject - Verb - Place - Time. Use this order to solve jumbled sentences quickly.

Q16. Correct answer: (B) have completed

'Each' is singular and needs a singular verb: 'Each of the boys has completed...'. The words 'of the boys' do not make the verb plural. 'Their' with 'each' is accepted in modern usage, so it is not the error the examiner wants.

Q17. Correct answer: (A) He does not know the answer.

With a third-person singular subject, 'does' is used and the main verb stays in its base form: does + not + know. The error 'does not knows' (double marking) is a classic trap. 'He not know' lacks an auxiliary verb.

Q18. Correct answer: (B) The proposal was approved by the committee.

Passive formula: object + form of 'be' + past participle + by + subject. The original verb is simple past, so the passive is 'was approved'. The last option reverses the meaning. Keep the same tense when changing voice.

Q19. Correct answer: (C) He said that he was busy.

In indirect speech, the reporting verb 'said' is past, so the tense shifts back (am becomes was) and the pronoun changes from 'I' to 'he'. 'Will be' changes the meaning to the future. Rule: present becomes past, past becomes past perfect.

Q20. Correct answer: (C) The news is good.

'News' looks plural because of the 's' but is an uncountable singular noun, so it takes 'is'. Other words in this family: mathematics, physics, economics, politics, measles. Trap: 'newses' does not exist.

PART-II: PROFESSIONAL TEST (80 Marks)

OSI Model and Network Security

Q21. Correct answer: (A) Network layer

The Network layer (Layer 3) uses logical (IP) addresses and routers to move packets between networks. The Data Link layer handles physical (MAC) addressing within one network, and the Transport layer provides end-to-end delivery. Memory aid: Layer 3 = Network = IP = Router = Packet.

Q22. Correct answer: (D) Presentation

The Presentation layer (Layer 6) prepares data for the application: encoding, encryption and compression. The Session layer (5) manages dialogs between systems. Trap: modern protocols like TLS work across layers, but in the textbook OSI answer encryption belongs to the Presentation layer.

Q23. Correct answer: (A) Segment

PDU names by layer: Application/Presentation/Session = Data, Transport = Segment (TCP) or Datagram (UDP), Network = Packet, Data Link = Frame, Physical = Bits. Memory aid (top to bottom): Data, Segment, Packet, Frame, Bit.

Q24. Correct answer: (B) UDP

UDP sends datagrams without setting up a connection, so it is fast but unreliable (used for streaming, DNS, VoIP). TCP is connection-oriented (three-way handshake, acknowledgements). HTTP and FTP are application protocols that run on top of TCP.

Q25. Correct answer: (D) Switch

A switch learns MAC addresses and sends frames only to the destination port. A hub and a repeater work at the Physical layer and simply regenerate or broadcast signals. A router works at the Network layer using IP addresses.

Q26. Correct answer: (B) Filter incoming and outgoing network traffic according to security rules

A firewall (hardware or software) allows or blocks traffic based on rules such as IP address, port and protocol. It does not improve speed or create backups. Converting names into IP addresses is the job of DNS.

Q27. Correct answer: (D) A pair of mathematically related public and private keys

Asymmetric encryption uses a public key (shared openly) and a private key (kept secret). Data encrypted with one can only be decrypted with the other. A single shared key describes symmetric encryption (AES, DES), which is faster but has a key-distribution problem.

Q28. Correct answer: (C) DDoS attack

Distributed Denial of Service (DDoS) overloads a server with requests from many sources (often a botnet). Phishing tricks users into revealing information, SQL injection attacks databases through input fields, and man-in-the-middle intercepts communication.

Q29. Correct answer: (C) Integrity

CIA = Confidentiality (only authorized people can read), Integrity (data stays accurate and unaltered), Availability (data is accessible when needed). Authentication is a security process but it is NOT one of the three CIA letters, which makes it a favourite trap.

Q30. Correct answer: (C) SHA-256

SHA-256 is a one-way hash function used for checking integrity and storing passwords. RSA is asymmetric encryption, while AES and DES are symmetric encryption algorithms. Key difference: hashing cannot be reversed, encryption can be decrypted with a key.

Fundamental Programming

Q31. Correct answer: (A) Compiler

A compiler translates the whole program at once and reports all errors together (C, C++). An interpreter translates and runs one line at a time (traditional Python behaviour). An assembler converts assembly language to machine code, and a loader places the program in memory.

Q32. Correct answer: (A) Syntax error

Syntax errors (missing semicolon, wrong keyword) are caught at compile time and stop the program from running. Logical errors give wrong output without any warning, and runtime errors occur while the program runs (such as division by zero).

Q33. Correct answer: (B) A named memory location whose value can change

A variable is a named storage location in memory. A fixed value is a constant, and reserved words (int, while) are keywords. Do not mix up variable (can change) and constant (cannot change).

Q34. Correct answer: (C) if-else

for, while and do-while repeat a block of code. if-else is a selection (decision) structure that chooses one path. The three basic control structures are sequence, selection and iteration.

Q35. Correct answer: (C) An informal, language-independent description of an algorithm

Pseudocode describes the steps of an algorithm in plain, structured English without strict syntax. A graphical representation of the same logic is a flowchart. Machine code is the binary output produced after compilation.

Data Types

Q36. Correct answer: (B) 4 bytes

The C++ standard only guarantees that int is at least 16 bits, but on most modern compilers it is 32 bits = 4 bytes. char is 1 byte, short is usually 2 bytes, and long long is 8 bytes. Use sizeof() to check on a given system.

Q37. Correct answer: (A) bool

bool holds one of two values: true (1) or false (0). char stores a single character, float stores decimal numbers, and string stores text.

Q38. Correct answer: (B) 3

Dividing two integers gives an integer result and the fractional part is discarded (integer division). To get 3.5 at least one operand must be a decimal, for example 7.0 / 2. The result is not rounded to 4. The remainder 1 can be obtained with 7 % 2.

Q39. Correct answer: (D) 9

Casting a decimal number to int truncates (cuts off) the fractional part; it does NOT round. So (int)9.99 = 9 and (int)-9.99 = -9. Truncation is a very common FPSC trap, because many candidates assume rounding.

Program Logic

Q40. Correct answer: (A) Diamond

A diamond shows a yes/no (true/false) decision with branching paths. A rectangle is a process, an oval (terminator) marks start/end, and a parallelogram represents input/output.

Q41. Correct answer: (A) False

5 > 3 is true, but 2 > 4 is false. AND gives true only when BOTH conditions are true, so the result is false. With OR the result would have been true, because only one true condition is enough.

Q42. Correct answer: (B) The two inputs are different

XOR outputs 1 only when exactly one input is 1 (0,1 or 1,0). When both inputs are equal (0,0 or 1,1) the output is 0. XOR is often described as the 'difference detector'.

Q43. Correct answer: (A) 10

The loop runs for i = 1, 2, 3 and 4, so sum = 1 + 2 + 3 + 4 = 10. The value 6 would result if the loop stopped at 3, and 15 if it ran up to 5. Always check whether the loop condition uses < or <=.

Basic Algorithms

Q44. Correct answer: (A) O(log n)

Binary search halves the search range at every step, so the number of steps is about log base 2 of n. O(n) belongs to linear search, and O(n log n) to efficient sorting algorithms such as merge sort.

Q45. Correct answer: (D) Bubble sort

Bubble sort compares neighbouring pairs, and the largest value 'bubbles' to the end in each pass. Selection sort picks the minimum and places it at the front, insertion sort inserts each element into a sorted portion, and quick sort partitions around a pivot.

Q46. Correct answer: (C) A sorted list

Binary search decides which half to discard by comparing with the middle element, which only works if the data is sorted. Duplicates and negative numbers are not a problem. For unsorted data, linear search is needed (or sort first).

Q47. Correct answer: (D) Stack

A stack removes the most recently added item first, like a pile of plates (push and pop). A queue follows FIFO (first in, first out), like people waiting in line (enqueue and dequeue).

Q48. Correct answer: (B) O(n)

In the worst case the target is at the last position or absent, so all n elements must be checked: O(n). The best case is O(1) when the element is first. O(log n) is for binary search on sorted data.

Computer Networks and Internet

Q49. Correct answer: (B) 1 to 126

Class A: 1-126, Class B: 128-191, Class C: 192-223, Class D (multicast): 224-239, Class E (experimental): 240-255. The value 127 is reserved for loopback (127.0.0.1), and some books write the range as 0-127; in that case choose the nearest option.

Q50. Correct answer: (C) Translate domain names into IP addresses

DNS works like the internet's phone book: it converts names such as example.com into numeric IP addresses. Automatic IP assignment is the job of DHCP, encryption of web traffic is done by TLS/HTTPS, and routing is done by routers.

Q51. Correct answer: (B) A central hub or switch

In a star topology every node has its own cable to a central device, so the failure of one cable affects only one node (but failure of the central device brings down the network). A single backbone cable is a bus topology, a closed loop is a ring, and direct links to everyone is a mesh.

Q52. Correct answer: (B) DHCP

DHCP (Dynamic Host Configuration Protocol) leases an IP address, subnet mask, gateway and DNS server to a device. ARP maps an IP address to a MAC address, and ICMP is used for error messages and tools like ping.

Q53. Correct answer: (D) 128 bits

IPv6 uses 128-bit addresses written in hexadecimal (for example 2001:db8::1), which solves the shortage of IPv4. IPv4 uses 32 bits. Remember: IPv4 = 32 bits, IPv6 = 128 bits.

Q54. Correct answer: (A) SMTP

SMTP (Simple Mail Transfer Protocol, port 25/587) sends and relays mail. POP3 and IMAP are used for retrieving mail from the server, and FTP transfers files. Memory aid: SMTP = Send, POP/IMAP = Pick up.

Q55. Correct answer: (A) MAN

MAN (Metropolitan Area Network) spans a city. A LAN covers a small area such as a building, a WAN covers countries or continents (the internet is the largest WAN), and a PAN covers a person's immediate surroundings (for example Bluetooth devices).

Q56. Correct answer: (B) /24

Each 255 represents 8 one-bits, so three 255s mean 24 network bits = /24. 255.0.0.0 is /8, 255.255.0.0 is /16, and /32 identifies a single host. A /24 network has 254 usable host addresses.

Designing, Solution and Testing

Q57. Correct answer: (B) Requirement analysis

The SDLC begins by understanding what the user needs (requirement gathering/analysis, sometimes preceded by planning). Design comes next, then coding, testing, deployment and maintenance. Coding before requirements is the classic mistake the examiner wants you to avoid.

Q58. Correct answer: (A) Each phase must be completed before the next phase begins

Waterfall is a linear sequential model: requirements, design, implementation, testing, deployment, maintenance, with no going back easily. Short repeating cycles and sprints describe Agile. Because changes are costly, Waterfall suits projects with stable, well-defined requirements.

Q59. Correct answer: (C) Black-box testing

Black-box testing checks inputs and outputs against requirements without looking at the code. White-box (structural, glass-box) testing needs knowledge of the code, and code coverage is a white-box measure.

Q60. Correct answer: (B) Unit testing

Unit testing is the lowest level and checks the smallest parts of a program. Integration testing checks that modules work together, system testing checks the complete system, and acceptance testing is done by the customer. Order: Unit, Integration, System, Acceptance.

Q61. Correct answer: (C) User acceptance testing (UAT)

UAT is the final testing stage before release and is performed by the client or end users in a real-world setting. Unit and integration testing are performed by developers/testers, and stress testing checks behaviour under extreme load.

Q62. Correct answer: (D) Use case diagram

A use case diagram shows actors (users or external systems) and the use cases (functions) they perform. A class diagram shows classes and relationships, and a sequence diagram shows the order of messages between objects over time.

Q63. Correct answer: (A) Confirm that new changes have not broken existing functionality

After a code change or bug fix, regression testing re-runs old tests to make sure previously working features still work. Heavy-load testing is performance/stress testing. Checking only new features misses the regression risk, which is the whole point.

Programming Languages: C++, Visual Basic and Python

Q64. Correct answer: (C) -> (arrow operator)

If p is a pointer, p->name is the same as (*p).name. The dot operator is used with an object itself, :: accesses class or namespace scope, and & gives the address of a variable. Rule: object uses dot, pointer uses arrow.

Q65. Correct answer: (B) // This is a comment

C++ uses // for single-line and /* ... */ for multi-line comments. # starts a comment in Python, an apostrophe starts a comment in Visual Basic, and <!-- --> is an HTML comment.

Q66. Correct answer: (A) 56

x++ is post-increment: the current value (5) is used first, then x becomes 6. The first cout prints 5, and the second prints 6, so the output is 56 (no space). With ++x, the output would be 66.

Q67. Correct answer: (D) Encapsulation

Encapsulation binds data and methods together and controls access using private/public. Inheritance lets one class acquire the properties of another, polymorphism allows one interface with many forms, and recursion is a function calling itself.

Q68. Correct answer: (A) It has the same name as the class and has no return type

A constructor is a special member function that runs automatically when an object is created and initializes it. It has no return type (not even void). Destroying an object is the job of the destructor, written with a tilde (~ClassName).

Q69. Correct answer: (C) new

new allocates memory on the heap at run time (int *p = new int;). delete releases that memory, sizeof gives the size of a type, and this is a pointer to the current object. Every new must have a matching delete to avoid memory leaks.

Q70. Correct answer: (D) &

The ampersand (&) is the dedicated concatenation operator in Visual Basic, for example "Hello " & "World". The + operator can also join two strings but may cause errors when mixing numbers and text, so & is preferred.

Q71. Correct answer: (B) Dim

Visual Basic declares variables with Dim (short for dimension), for example Dim age As Integer. Var and Let belong to JavaScript, and Int is a type in C-style languages, not a declaration keyword in VB.

Q72. Correct answer: (D) MsgBox

MsgBox (or MessageBox.Show in VB.NET) shows a message window with buttons such as OK or Yes/No. Echo is a PHP statement, cout is used for console output in C++, and alert() belongs to JavaScript.

Q73. Correct answer: (A) <class 'list'>

Square brackets [] create a list, so type([]) shows <class 'list'>. A tuple uses (), a dictionary uses {key: value}, and Python has no built-in type named array (it exists in a separate module).

Q74. Correct answer: (D) tuple

A tuple cannot be changed after creation (no item assignment). Lists, dictionaries and sets can be modified. Other immutable types: int, float, str, bool, frozenset.

Q75. Correct answer: (C) def

Python functions are defined with def, for example: def add(a, b): return a + b. The keyword 'function' belongs to JavaScript, and 'fun' is used in Kotlin.

Q76. Correct answer: (D) 8

The ** operator means exponent, so 2 ** 3 = 2 x 2 x 2 = 8. The value 6 would come from multiplication (2 * 3), and 5 from addition. Do not confuse it with ^ which is bitwise XOR in Python.

Q77. Correct answer: (B) 3 1

// is floor division (integer quotient), so 10 // 3 = 3, and % gives the remainder, so 10 % 3 = 1. Normal division with / would give 3.333... Python print() separates multiple values with a space, so the result is: 3 1.

Unix / Red Hat Linux Operating System

Q78. Correct answer: (C) ls

ls lists directory contents (ls -l gives the long format and ls -a shows hidden files). cd changes directory, rm removes files, and cat displays file content.

Q79. Correct answer: (D) chmod

chmod (change mode) sets read, write and execute permissions. chown changes the owner of a file, chgrp changes the group, and passwd changes a password. Memory aid: chmod = mode, chown = owner.

Q80. Correct answer: (B) rwxr-xr-x

Each digit is the sum of r = 4, w = 2, x = 1. So 7 = rwx (owner), 5 = r-x (group) and 5 = r-x (others). 777 would be rwxrwxrwx, 644 is rw-r--r--, and 700 is rwx------.

Q81. Correct answer: (B) Linus Torvalds

Linus Torvalds released the first Linux kernel in 1991. Richard Stallman started the GNU project and the free-software movement, while Ken Thompson and Dennis Ritchie created the original UNIX at Bell Labs (Ritchie also created C).

Q82. Correct answer: (A) pwd

pwd stands for 'print working directory'. cd changes the directory, whoami shows the current user name, and hostname shows the system's name.

Q83. Correct answer: (A) .rpm

Red Hat, CentOS, Fedora and Rocky/Alma Linux use RPM packages, managed with rpm, yum or dnf. .deb is used by Debian and Ubuntu (apt/dpkg), .apk by Android and Alpine Linux, and .exe is a Windows executable.

Q84. Correct answer: (C) /root

The root user's home directory is /root. Ordinary users live under /home/username. /etc stores configuration files and /usr stores user programs. Trap: /home is the parent of ordinary users' directories, not root's.

Q85. Correct answer: (B) more

more (and its improved version less) shows the file page by page. cat prints the whole file at once, touch creates an empty file or updates its timestamp, and mkdir creates a directory.

Q86. Correct answer: (C) Search for text patterns inside files

grep searches files or command output for lines that match a pattern, for example grep "error" log.txt. Compression is done with tar, gzip or zip, ownership with chown, and processes are listed with ps or top.

Multiple Browsers

Q87. Correct answer: (B) Gecko

Firefox uses Gecko. Blink powers Chrome, Edge and Opera, WebKit powers Safari, and Trident was the engine of the old Internet Explorer. Matching browser to engine is a common one-mark question.

Q88. Correct answer: (D) It does not save history, cookies and form data on the device after the session ends

Private mode only avoids storing local traces on that device. Websites, your employer and your ISP can still see your activity and IP address. Hiding the IP needs a VPN or proxy. Trap: many candidates believe incognito means fully anonymous.

Q89. Correct answer: (A) Apache

Apache HTTP Server is web server software that delivers web pages. A browser is client software that requests and displays them. Opera, Safari and Edge are all browsers.

Search Engine Optimization (SEO)

Q90. Correct answer: (B) Improve a website's visibility in organic (unpaid) search results

SEO aims to rank higher in unpaid results so that more visitors arrive naturally. Paying for ad placement is SEM/PPC advertising (for example Google Ads). Speed and design are only parts of SEO, not its main goal.

Q91. Correct answer: (B) Optimizing title tags and meta descriptions

On-page SEO covers things you control on your own pages: titles, meta descriptions, headings, content quality, URLs, image alt text. Backlinks, social sharing and guest posts happen outside your site, so they are off-page SEO.

Q92. Correct answer: (A) Links from other websites that point to your website

A backlink (inbound link) is a link on another site pointing to yours. Search engines treat quality backlinks as votes of trust. Links within the same website are internal links.

Q93. Correct answer: (B) Tell search engine crawlers which parts of a site they should not crawl

robots.txt sits in the site's root folder and gives instructions to crawlers. Trap: it is a request, not security, so it must never be used to hide private data. A sitemap.xml file does the opposite job by listing pages you want crawled.

Q94. Correct answer: (D) Keyword stuffing

Black-hat SEO uses tricks that violate search engine guidelines, such as keyword stuffing, hidden text and buying links, and risks a penalty. The other three are white-hat practices that follow the guidelines.

Web Development

Q95. Correct answer: (D) <h1>

Headings run from <h1> (largest) to <h6> (smallest). <head> is not a heading; it holds page information such as metadata, and <title> sets the text on the browser tab. Trap: <head> looks like a heading but is not.

Q96. Correct answer: (D) color

In CSS the property is simply color (for example p { color: blue; }). font-color and text-color do not exist. To change the background use background-color.

Q97. Correct answer: (C) <a>

The anchor tag <a href="..."> creates a link. href is an attribute, not a tag. <link> is placed inside <head> to attach resources such as stylesheets, and <url> does not exist in HTML.

Q98. Correct answer: (A) JavaScript

JavaScript normally runs inside the user's browser. PHP, ASP and JSP run on the server and send the finished HTML to the browser. Note: with Node.js, JavaScript can also run on the server, but the standard exam answer is client-side.

Q99. Correct answer: (C) Not Found

404 means the server could not find the requested page or resource. Other common codes: 200 = OK, 403 = Forbidden, 500 = Internal Server Error, 301 = Moved Permanently. Rule: 4xx = client error, 5xx = server error.

Q100. Correct answer: (C) POST

POST sends data in the request body, so it is suitable for sensitive or large data such as passwords and file uploads. GET appends data to the URL, where it is visible in history and logs. HEAD asks only for headers, and TRACE is used for diagnostics.

Score Calculation and Interpretation

Score = (number correct) - 0.25 x (number wrong). Unattempted questions count as zero. Example: 70 correct, 20 wrong, 10 skipped gives 70 - 5 = 65 marks.

Score (out of 100)	Result	What to do next
Below 40	Below passing level	Re-read the notes of every topic, then retake this test after a few days.
40 - 54	Borderline pass	One careless mistake can fail you. Fix wrong answers and avoid guessing.
55 - 69	Good	Competitive for many posts. Aim for accuracy in the topics you missed.
70 - 84	Very good	Strong preparation. Practice with another full mock under timed conditions.
85 - 100	Excellent	Keep revising regularly and maintain speed and accuracy.

Calculate your score here: Correct = _____ Wrong = _____ Final = Correct - (0.25 x Wrong) = _____